THE SCALABLE QUANTUM MANY BODY SPIN SYSTEMS USING VARIATINAL APPRAOACH ON TFIM

The one-dimensional transverse Ising model (TFIM) describes a chain of spin-½ particles arranged in a line, where:

Each spin interacts with its nearest neighbor along the z-direction.

An external magnetic field is applied in the transverse (x) direction.

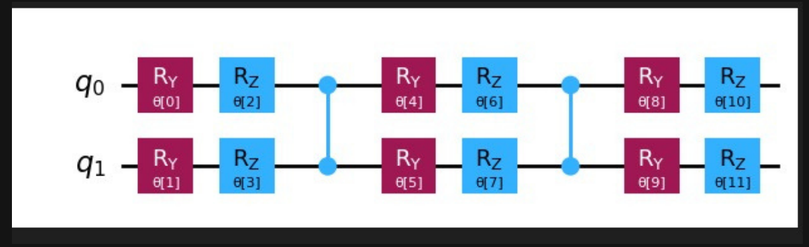
Approaches :

- VQE AND VQD
- ML

Explanation about VQE

Variational Quantum Eigensolver (VQE) is a hybrid quantum–classical algorithm used to find the ground state energy of a Hamiltonian.

- OPERATOR
- ANSATZ
- EESTMATOR
- CLASSICAL OPTIMIZER



Classical computation

Exact diagonalization

Construct full Hamiltonian matrix

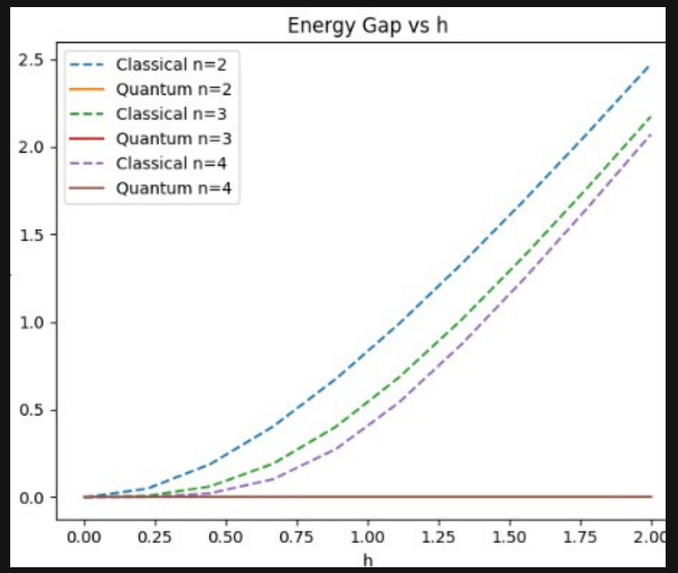
Diagonalize numerically

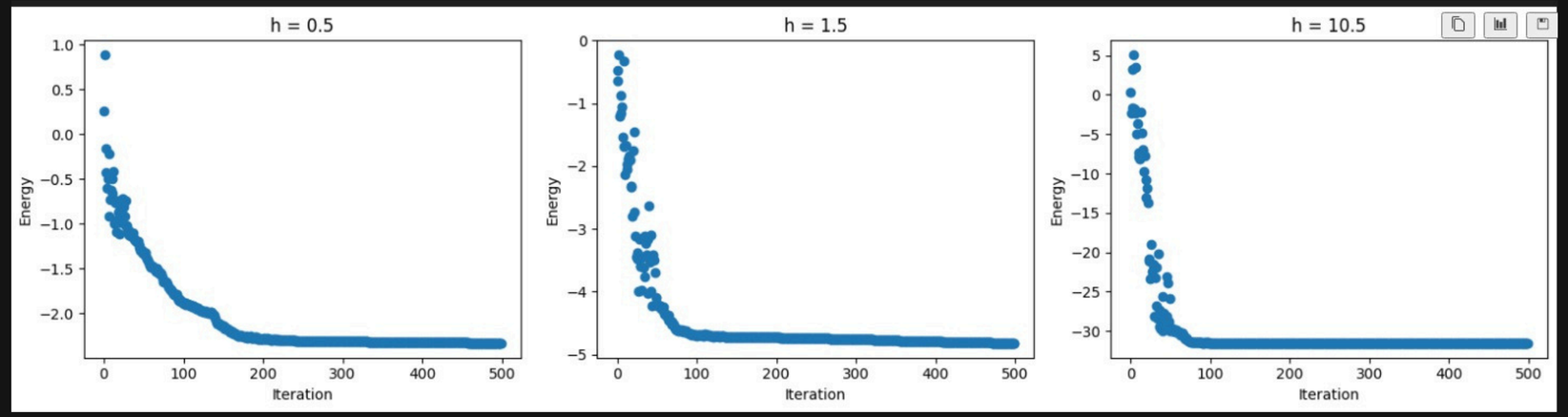
Works only for small N

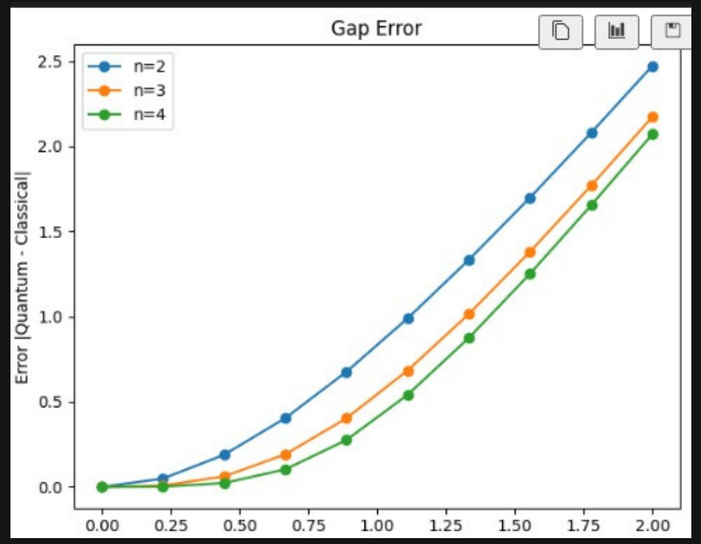
```python
n = 4
J = 1.0
h_values = np.linspace(2,3,10)
gaps_classical = []
gaps_quantum = []
gaps_error = []
for h in h_values:
    H = build_tfim(n, J, h)
    E0_Classical, E1_classical, gap_c = classical_energy(H)
    gaps_classical.append(gap_c)
    E0 = compute_ground_energy(H, n)
    E1 = compute_excited_energy(H, n)
    gaps_quantum.append(E1 - E0)
    gaps_error.append(np.abs(E1 - E0 - gap_c))
    print(f"h={h:.2f} | Classical:gap={gap_c:.4f};Ground={E0_Classical:.4f};excited={E1_classical:.4f} | Quantum:gap={E1-E0:.4f};Ground={E0:.4f};excited={E1:.4f}")
```

✓ 7m 40.0s Python

```
h=2.00 | Classical:gap=2.5110;Ground=-8.3768;excited=-5.8658 | Quantum:gap=4.1185;Ground=-8.2129;excited=-4.0944
h=2.11 | Classical:gap=2.7250;Ground=-8.8013;excited=-6.0762 | Quantum:gap=3.3809;Ground=-8.6381;excited=-5.2572
h=2.22 | Classical:gap=2.9400;Ground=-9.2277;excited=-6.2877 | Quantum:gap=3.6309;Ground=-9.0768;excited=-5.4459
h=2.33 | Classical:gap=3.1558;Ground=-9.6559;excited=-6.5001 | Quantum:gap=4.4993;Ground=-9.3787;excited=-4.8794
h=2.44 | Classical:gap=3.3723;Ground=-10.0856;excited=-6.7133 | Quantum:gap=4.8066;Ground=-9.9714;excited=-5.1648
h=2.56 | Classical:gap=3.5894;Ground=-10.5166;excited=-6.9272 | Quantum:gap=4.1951;Ground=-10.4427;excited=-6.2476
h=2.67 | Classical:gap=3.8069;Ground=-10.9487;excited=-7.1418 | Quantum:gap=4.3444;Ground=-10.6882;excited=-6.3437
h=2.78 | Classical:gap=4.0249;Ground=-11.3818;excited=-7.3569 | Quantum:gap=5.1454;Ground=-11.3548;excited=-6.2094
h=2.89 | Classical:gap=4.2433;Ground=-11.8158;excited=-7.5725 | Quantum:gap=5.2724;Ground=-11.5879;excited=-6.3155
h=3.00 | Classical:gap=4.4619;Ground=-12.2506;excited=-7.7886 | Quantum:gap=6.1171;Ground=-12.1122;excited=-5.9952
```







Prominant Limitations :

- Ansatz dependencies

When we use VQE to find TFIM groundstate we choose parameterized wave function (Ansatz)

If Ansatz is too simple it cannot represent the ground state .Energy never converges properly

- Scaling Issues

Number of quiet is limitted . Connectivity between quibit is limitted . Circuit depth increases with system state

More spin – more entanglement
More entanglement – Deeper Circuits
Deep Circuits– more noise so scaling become difficult

t, r, θ, ϕ

$$g_{\mu\nu} = \begin{bmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{10} & g_{11} & g_{12} & g_{13} \\ g_{20} & g_{21} & g_{22} & g_{23} \\ g_{30} & g_{31} & g_{32} & g_{33} \end{bmatrix}$$

Let assume it to be static and spherical symmetric

$$g_{\mu\nu} = \begin{bmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{01} & g_{11} & g_{21} & g_{31} \\ g_{02} & g_{12} & r^2 & 0 \\ g_{03} & g_{13} & 0 & r^2\sin^2\theta \end{bmatrix}$$

Let

$t \to -t$ $\quad g_{01} = g_{\alpha\beta}\frac{\partial x^\alpha}{\partial x'^0}\frac{\partial x^\beta}{\partial x'^1} = -g_{01} \Rightarrow g_{01} = 0$, $g_{02} = 0$, $g_{03} = 0$

$\theta \to -\theta$ $\quad g_{12} = 0$

$\phi \to -\phi$ $\quad g_{13} = 0$

$$\therefore g_{\mu\nu} = \begin{bmatrix} g_{00} & 0 & 0 & 0 \\ 0 & g_{11} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix}$$

Let $g_{00} = -A(r)$, $g_{11} = B(r)$

Mass only $\Rightarrow R_{\mu\nu}$ outside $\Rightarrow 0$

$R_{\mu\nu} = 0$

$$\partial_\lambda \Gamma^\lambda_{\mu\nu} - \partial_\nu \Gamma^\lambda_{\mu\lambda} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{\mu\nu} - \Gamma^\lambda_{\nu\sigma}\Gamma^\sigma_{\mu\lambda} = 0$$

$$\therefore \Gamma^i_{jk} = \frac{1}{2} g^{i\sigma}(\partial_j g_{\sigma k} + \partial_k g_{j\sigma} - \partial_\sigma g_{jk})$$

$$\Gamma^r_{tt} = \frac{1}{2} g^{rr}(-\partial_r g_{tt}) = \frac{A'(r)}{2B(r)}$$

$$\Gamma^t_{tr} = \frac{1}{2} g^{tt}(\partial_r g_{tt}) = \frac{A'(r)}{2A(r)} = \Gamma^t_{rt}$$

$$\Gamma^r_{rr} = \frac{1}{2} g^{rr}(\partial_r g_{rr} + \partial_r g_{rr} - \partial_r g_{rr}) = \frac{B'}{2B}$$

$$\Gamma^r_{r\theta} = \frac{1}{2} g^{rr}(0) = 0 \qquad \Gamma^r_{r\phi} = \frac{1}{2} g^{rr}(0) = 0$$

$$\Gamma^\phi_{r\phi} = \frac{1}{2} g^{\phi\phi}(\partial_r g_{\phi\phi}) = \frac{1}{2r^2\sin^2\theta} \times 2r\sin^2\theta = \frac{1}{r} = \Gamma^\phi_{\phi r}$$

$$\Gamma^\theta_{r\theta} = \frac{1}{2} g^{\theta\theta}(\partial_r g_{\theta\theta}) = \frac{1}{2r^2} \times 2r = \frac{1}{r} = \Gamma^\theta_{\theta r}$$

$$\Gamma^r_{\theta\theta} = \frac{1}{2} g^{rr}(-\partial_r g_{\theta\theta}) = \frac{1}{2B}(-2r) = -\frac{r}{B}$$

$$\Gamma^r_{\phi\phi} = \frac{1}{2} g^{rr}(-\partial_r g_{\phi\phi}) = \frac{1}{2B}(-2r\sin^2\theta) = -\frac{r\sin^2\theta}{B}$$

$$\Gamma^\theta_{\phi\phi} = \frac{1}{2} g^{\theta\theta}(-\partial_\theta g_{\phi\phi}) = \frac{1}{2r^2} \times -r^2 \cdot 2\sin\theta\cos\theta = -\sin\theta\cos\theta$$

$$\Gamma^\phi_{\theta\theta} = \frac{1}{2} g^{\phi\phi}(-\partial_\phi g_{\theta\theta}) = 0$$

As A Function $\Rightarrow R_{\mu\nu} = 0 \Rightarrow R_{tt} = R_{rr} = R_{\theta\theta} = R_{\phi\phi} = 0$

$$\therefore R_{tt} = \partial_\lambda \Gamma^\lambda_{tt} - \underbrace{\partial_t \Gamma^\lambda_{t\lambda}}_{\to 0 \text{ (nothing changes with time)}} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{tt} - \Gamma^\lambda_{t\sigma}\Gamma^\sigma_{\lambda t}$$

(i) $\partial_\lambda \Gamma^\lambda_{tt} = \partial_r \Gamma^r_{tt} = \partial_r\left(\frac{A'}{2B}\right) = \frac{A''B - A'B'}{2B^2} = \frac{A''}{2B} - \frac{A'B'}{2B^2}$

(ii) $\partial_t \Gamma = 0$

(iii) $\Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{tt} = \left(\Gamma^t_{tr} + \Gamma^r_{rr} + \Gamma^\theta_{\theta r} + \Gamma^\phi_{\phi r}\right)\Gamma^r_{tt}$

$$\Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{tt} = \frac{(A')^2}{4AB} + \frac{A'B'}{2B^2} + \frac{A'}{Br}$$

(iv) $\Gamma^\lambda_{t\sigma}\Gamma^\sigma_{\lambda t} = \Gamma^t_{tr}\Gamma^t_{rt} + \Gamma^r_{rr}\Gamma^r_{rr} + \Gamma^\theta_{\theta r}\Gamma^\theta_{r\theta} + \Gamma^\phi_{\phi r}\Gamma^\phi_{r\phi}$

$$= \frac{(A')^2}{4A^2} + \frac{(B')^2}{2B^2} + \frac{2}{r^2}$$

$$R_{rr} = \frac{-A''}{2A} + \frac{(A')^2}{4A^2} + \frac{A'B'}{4AB} + \frac{B'}{Br} = 0$$

$$R_{\lambda\lambda} = -\frac{A''}{2A} + \frac{A'}{4A}\left(\frac{A'}{A} + \frac{B'}{B}\right) + \frac{B'}{Br} = 0 \to ②$$

① $\Rightarrow$ ~~$-2AA''$~~ ~~$+(A')^2$~~ $+BA'A' - AA'B' = 0$

② $\Rightarrow$ ~~$-2ABA''$~~ ~~$+B(A')^2$~~ ~~$+AA'B'$~~ $+4A^2B' = 0$

① + ② $\Rightarrow 4A(AB' + BA') = 0$

$A \neq 0$ $\quad \partial_r(-AB) = 0$

$\therefore AB = \text{const.} \Rightarrow$

$$R_{rr} = \frac{A''}{2A} - \frac{A'}{4A}\left(\frac{A'}{A} + \frac{B'}{B}\right) + \frac{A'}{Br} = 0 \to ①$$

$$\therefore R_{\lambda\lambda} = \partial_\lambda \Gamma^\lambda_{\lambda\lambda} - \partial_\lambda \Gamma^\lambda_{\lambda\lambda} + \Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{\lambda\lambda} + \Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\lambda\lambda}$$

(i) $\partial_\lambda \Gamma^\lambda_{\lambda\lambda} = \partial_r \Gamma^r_{rr} = \frac{B''}{2B} - \frac{(B')^2}{2B^2}$

(ii) $\partial_\lambda \Gamma^\lambda_{\lambda\lambda} = \partial_\lambda\left[\Gamma^t_{\lambda t} + \Gamma^r_{\lambda r} + \Gamma^\theta_{\lambda\theta} + \Gamma^\phi_{\lambda\phi}\right] = \frac{A''}{2A} - \frac{(A')^2}{2A^2} + \frac{B''}{2B} - \frac{(B')^2}{2B^2} - \frac{2}{r^2}$

(iii) $\Gamma^\lambda_{\lambda\sigma}\Gamma^\sigma_{\lambda\lambda} = \left[\Gamma^t_{tr} + \Gamma^r_{rr} + \Gamma^\theta_{r\theta} + \Gamma^\phi_{r\phi}\right]\Gamma^r_{rr} = \frac{A'B'}{4AB} + \frac{(B')^2}{4B^2} + \frac{B'}{Br}$

(iv) $\Gamma^\lambda_{\sigma\lambda}\Gamma^\sigma_{\lambda\lambda} = \Gamma^t_{tr}\Gamma^t_{rt} + (\Gamma^r_{rr})^2 + (\Gamma^\theta_{\theta r})^2 + (\Gamma^\phi_{\phi r})^2$

$$= \frac{(A')^2}{4A^2} + \frac{(B')^2}{2B^2} + \frac{2}{r^2}$$

$$R_{rr} = \frac{-A''}{2A} + \frac{(A')^2}{4A^2} + \frac{A'B'}{4AB} + \frac{B'}{Br} = 0$$

$$R_{\lambda\lambda} = -\frac{A''}{2A} + \frac{A'}{4A}\left(\frac{A'}{A} + \frac{B'}{B}\right) + \frac{B'}{Br} = 0 \to ②$$

① $\Rightarrow$ ~~$-2AA''$~~ ~~$+(A')^2$~~ $+BA'A' - AA'B' = 0$

② $\Rightarrow$ ~~$-2ABA''$~~ ~~$+B(A')^2$~~ ~~$+AA'B'$~~ $+4A^2B' = 0$

① + ② $\Rightarrow 4A(AB' + BA') = 0$

$A \neq 0$ $\quad \partial_r(-AB) = 0$

$\therefore AB = \text{const.} \Rightarrow$

$B = \frac{1}{A}$ in eqn (i)

$$2LAA'' - LA(A')^2 + 4LA' + L(A')^2 = 0$$

$$A'' + 2\frac{A'}{r} = 0$$

$$A'' = -\frac{2A'}{r}$$

$$\frac{A''}{A'} = \frac{-2}{r} \Rightarrow \ln A' = -2\ln r$$

$$A' = \frac{C_1}{r^2} \Rightarrow A = C_1 - \frac{C_2}{r}$$

$$A = A_0 - \frac{k}{r}$$

$$\therefore g_{\mu\nu} = \begin{bmatrix} -(A_0 - \frac{k}{r}) & 0 & 0 & 0 \\ 0 & \frac{1}{A_0 - \frac{k}{r}} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix}$$

$\Rightarrow r \to \infty$, $g_{\mu\nu} = \text{flat}$, $\eta_{\mu\nu}$

$$g_{\mu\nu} = \begin{bmatrix} -A_0 & 0 & 0 & 0 \\ 0 & \frac{1}{A_0} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix} \qquad -A_0 = -1, \ \frac{1}{A_0} = 1 \to$$

$$\therefore g_{\mu\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{1}{1-\frac{k}{r}} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix}$$

$\Rightarrow \Gamma^r_{tt} = \frac{A'}{2B} = -\frac{k}{2r^2}\left(1 - \frac{k}{r}\right)$ $\quad -\frac{\partial g_{tt}}{\partial r} = \frac{k}{r^2}$, $\frac{\partial g_{tt}}{\partial t} = 0$

r changes based on $t \to$ $\Gamma^r_{tt} = \frac{\partial^2 r}{\partial t^2} = -\frac{k}{2r^2}$

$(r \to \infty)$ $\quad \frac{k}{2} = GM \Rightarrow$

$$\therefore g_{\mu\nu} = \begin{bmatrix} \frac{2GM}{r} - 1 & 0 & 0 & 0 \\ 0 & \frac{1}{1-\frac{2GM}{r}} & 0 & 0 \\ 0 & 0 & r^2 & 0 \\ 0 & 0 & 0 & r^2\sin^2\theta \end{bmatrix}$$

$$\boxed{\therefore ds^2 = \left(\frac{2GM}{r} - 1\right)dt^2 - \left(1 - \frac{2GM}{r}\right)^{-1}dr^2 + r^2 d\Omega^2}$$